Integrating Building Information modelling (BIM) and Artificial Intelligence (AI) for smart construction schedule, cost, quality, and safety management: challenges and opportunities

*1 Nitin Liladhar Rane

*1 University of Mumbai, Mumbai, India
Email: nitinrane33@gmail.com

Abstract:

In recent times, the construction industry has experienced notable progress, especially through the amalgamation of Building Information Modelling (BIM) and Artificial Intelligence (AI). This integration holds significant potential for transforming the management of construction schedules, costs, quality, and safety, thus amplifying the overall efficiency and efficacy of projects. Nonetheless, the merging of BIM and AI presents its own set of challenges, and comprehending these obstacles is crucial for effectively implementing and fostering this symbiotic relationship. This research critically evaluates the current challenges and future scope linked to the integration of BIM and AI for intelligent construction management. The primary challenges predominantly revolve around data integration and interoperability. BIM and AI systems frequently operate on different platforms and data structures, resulting in intricacies in data exchange and synchronization. Additionally, concerns related to data privacy and security pose substantial barriers to the utilization of AI algorithms on BIM data, prompting worries about safeguarding and potential misuse of sensitive project information. Furthermore, proficient utilization of AI in BIM for construction management necessitates a thorough understanding of AI algorithms, requiring comprehensive training and skill development among industry professionals. In this context, the research underscores the necessity for a comprehensive framework for training and education to facilitate the widespread adoption and implementation of AI-integrated BIM practices in the construction sector. The research also underscores the promising future prospects of this integration, highlighting the potential for improved decision-making processes, predictive analytics, and real-time monitoring capabilities. It stresses the significance of establishing standardized protocols and guidelines to ensure seamless integration and interoperability between BIM and AI systems. Furthermore, it emphasizes the importance of continuous research and development to address the existing challenges and unlock the full potential of this integrated approach for intelligent construction management.

Keywords: Building Information Modelling, Artificial Intelligence, Construction management, construction schedule, Cost, Quality, Safety management.

Introduction

The integration of Building Information Modelling (BIM) and Artificial Intelligence (AI) represents a significant leap forward for the construction sector, enabling the creation of intelligent solutions for efficient scheduling, cost management, quality assurance, and enhanced safety [1-6]. BIM, a sophisticated 3D modeling process, has transformed the construction landscape, empowering professionals to plan, design, construct, and manage structures and infrastructure effectively [7-9]. Simultaneously, AI's cognitive computing capabilities have augmented the industry's capacity to process complex data, leading to informed decision-making and fostering innovation in construction management [10-13]. The integration of BIM and AI offers a unique opportunity to revolutionize conventional construction practices into a data-driven, automated framework, effectively addressing persistent challenges such as project delays, cost overruns, quality variations, and safety risks [2,5,6]. By leveraging the collective potential of BIM and AI, construction stakeholders can streamline operations, boost productivity, reduce errors, and adhere to stringent safety standards, thereby promoting sustainable development and resource optimization in the construction sector.

Despite the potential promise of this integration, its implementation is riddled with various challenges that demand careful consideration. These obstacles include technical intricacies, interoperability issues, data management concerns, organizational preparedness, and the necessity for a comprehensive shift in the industry's mindset and practices. Overcoming these hurdles is crucial to harnessing the full potential of BIM and AI integration, allowing stakeholders to capitalize on the benefits of data-driven decision-making, automated processes, and optimized resource utilization. This research aims to comprehensively explore the challenges and future directions related to the integration of BIM and AI for smart construction management. Through an analysis of existing literature, empirical data, and insights, this study seeks to provide a nuanced understanding of the current landscape and potential pathways for the seamless integration of BIM and AI. Additionally, it endeavors to propose strategic recommendations and best practices to guide stakeholders in leveraging this integration's transformative potential for promoting sustainable, efficient, and safe construction practices. Figure 1 shows the co-occurrence analysis of the keywords in literature.

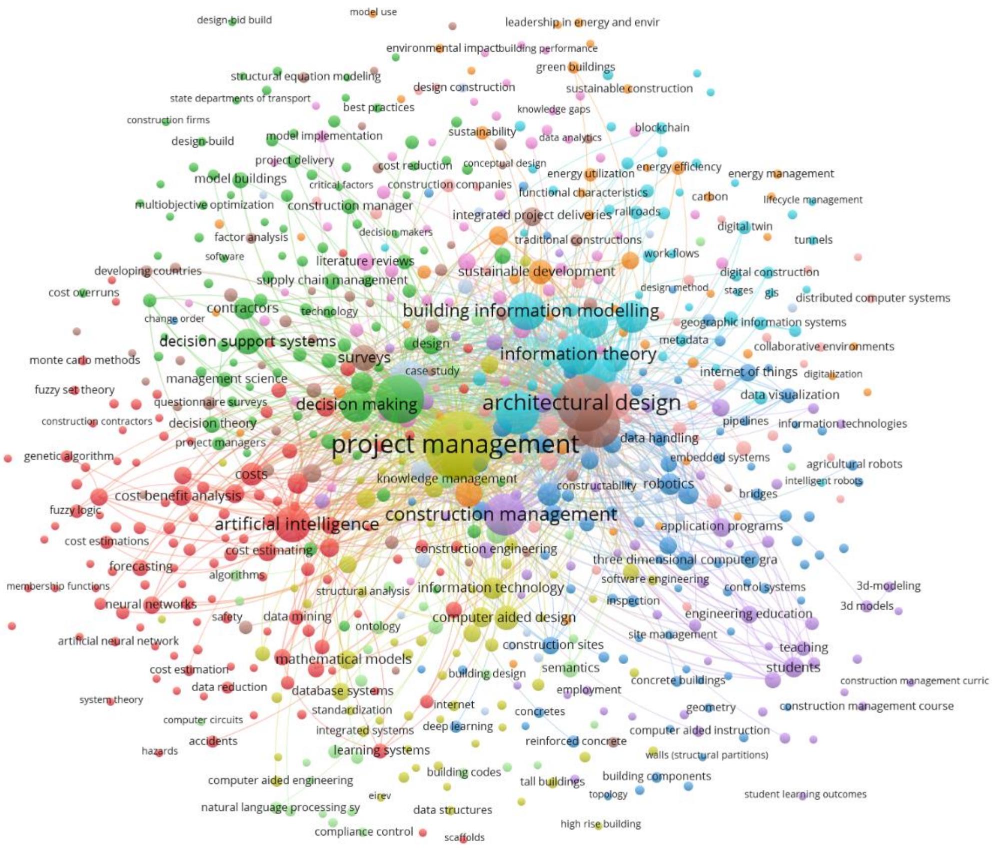

Figure 1 Co-occurrence analysis of the keywords in literature

Understanding the complex interplay between BIM and AI within the construction industry requires a multifaceted examination that considers not only technical aspects but also organizational and cultural dynamics [5,6]. Grasping these intricacies is crucial for devising effective strategies that can mitigate challenges and harness the full potential of BIM and AI integration to facilitate sustainable growth and advancement within the construction sector. This research contributes to the ongoing discourse on the transformative role of technology in the construction industry by shedding light on the challenges and opportunities inherent in the integration of BIM and AI. It emphasizes the significance of cultivating a collaborative ecosystem that encourages innovation, knowledge sharing, and the adoption of cutting-edge technologies to drive continuous improvement and competitiveness within the construction domain.

The subsequent sections of this research delve deeper into the complexities of the challenges and opportunities associated with the integration of BIM and AI in smart construction management. Through a comprehensive review of literature, the paper provides insights into the technical complexities and interoperability challenges that often hinder the seamless integration of BIM and AI. It also highlights the importance of establishing robust data management protocols and fostering a culture of innovation and collaboration to ensure the successful implementation of BIM and AI initiatives in construction projects. Moreover, the research emphasizes the need for tailored strategies that align with the specific requirements and objectives of construction stakeholders, underscoring the significance of a holistic approach that considers both the technical and human aspects of integration.

By synthesizing key findings and drawing from practical experiences and expert insights, this research aims to provide actionable recommendations and guidelines for industry stakeholders to effectively navigate the complexities of BIM and AI integration. These recommendations are tailored to address challenges related to data standardization, interoperability, skill development, and organizational change management, enabling stakeholders to maximize the benefits of BIM and AI integration while mitigating potential risks and challenges. This research underscores the transformative potential of integrating BIM and AI in the construction industry, providing a roadmap for stakeholders to overcome challenges and capitalize on the myriad opportunities presented by this synergy. By fostering a culture of collaboration, innovation, and adaptability, the construction sector can leverage the full potential of BIM and AI to drive sustainable development, enhance project efficiency, and ensure the delivery of high-quality, safe, and cost-effective construction projects in the years to come [4,6]. Figure 2 shows the co-authorship analysis.

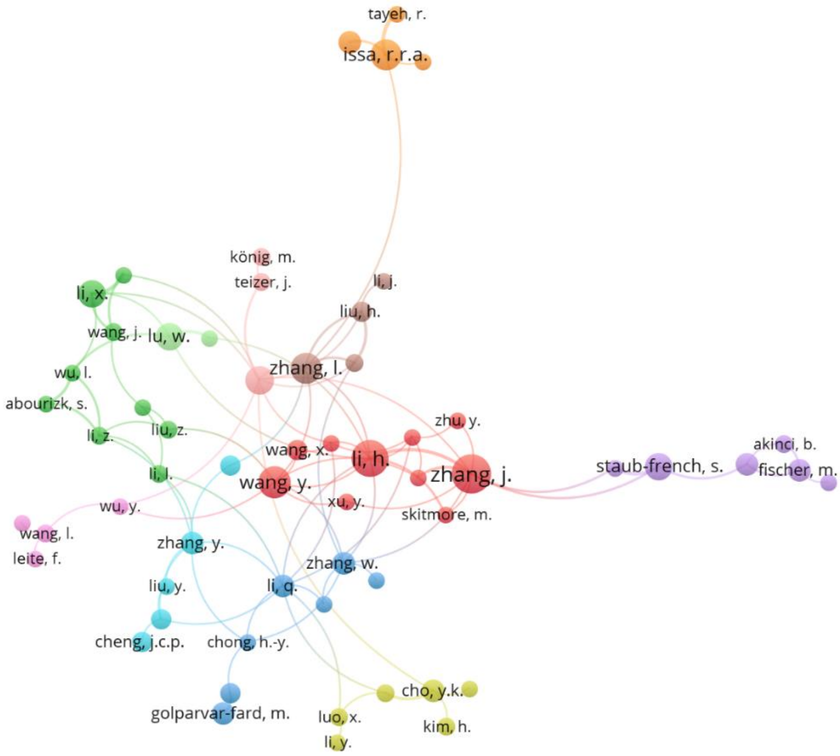

Figure 2 Co-authorship analysis

Integrating BIM and AI for smart construction schedule management

The fusion of Building Information Modelling (BIM) and Artificial Intelligence (AI) in construction scheduling signifies a groundbreaking approach within the construction sector, introducing unprecedented levels of efficiency, precision, and optimization to project management [14-20]. BIM serves as a digital representation of both the physical and functional aspects of a building, while AI, particularly through machine learning algorithms, interprets this data to make informed decisions. When these technologies are combined, construction scheduling becomes more streamlined, cost-effective, and adaptable to dynamic project conditions [16,19]. One of the primary advantages of integrating BIM and AI into construction scheduling is the enhanced visualization and coordination capabilities [15,19]. BIM provides a comprehensive 3D model of the entire project, encompassing all its components and systems [8,9]. AI algorithms can analyze this data to detect potential clashes or conflicts in the design, enabling early issue resolution [11,13]. This proactive approach minimizes rework, reduces delays, and ensures a smoother construction process.

Furthermore, AI algorithms can analyze historical project data and real-time information to anticipate potential risks and delays [11,13]. Through machine learning techniques, construction teams can foresee challenges and allocate resources more effectively. For instance, AI can analyze past project schedules, taking into account factors like weather conditions, resource availability, and team productivity, to create more precise and dependable schedules for future projects. This predictive analysis assists construction managers in making informed decisions, adjusting schedules in real-time, and optimizing resource allocation, thereby minimizing project delays and costs. Additionally, the integration of BIM and AI enables the automation of various tasks related to construction scheduling. AI algorithms can automate schedule generation based on project requirements, resource availability, and constraints. These automated schedules can be continually optimized as new data becomes available, ensuring that the construction process remains on track and within budget. Automation also reduces the administrative workload on project managers, allowing them to concentrate on more strategic aspects of project management [5,11,12].

Another significant advantage of smart construction scheduling through BIM and AI integration is the ability to conduct "what-if" analyses. Construction projects often encounter uncertainties, such as weather disruptions, material shortages, or unexpected design changes. BIM provides a comprehensive digital model that can be manipulated and analyzed in various scenarios [8,9]. AI algorithms can simulate different conditions and assess their impact on the project schedule. This capability empowers construction teams to devise contingency plans and make proactive decisions to mitigate potential risks [10,11]. By exploring different scenarios, project managers can identify the most efficient and cost-effective scheduling strategies, ensuring that the project stays on course even in the face of unforeseen challenges. Moreover, BIM and AI integration fosters real-time collaboration and communication among project stakeholders. The digital nature of BIM allows all parties involved in the project, including architects, engineers, contractors, and clients, to access the same up-to-date information. AI algorithms can process this shared data and provide valuable insights to each stakeholder, enabling better-informed decision-making. For example, construction managers can receive automated alerts and recommendations from AI systems, allowing them to address issues promptly and keep the project on schedule. This real-time collaboration promotes a more transparent and efficient workflow, enhancing overall project coordination and communication. Table 1 shows the integration of Building Information modelling (BIM) and Artificial Intelligence (AI) for construction schedule, cost, quality, and safety management.

Furthermore, the integration of BIM and AI in construction scheduling promotes sustainable practices and resource optimization [15,18]. AI algorithms can analyze data related to material usage, energy consumption, and waste generation, providing valuable insights into sustainable construction practices. By optimizing construction schedules based on these insights, projects can reduce their environmental impact and promote eco-friendly building practices. Additionally, AI can help optimize the use of resources such as manpower, equipment, and materials, minimizing waste and maximizing efficiency [11,12]. This resource optimization not only reduces costs but also contributes to the overall sustainability of the construction project. Moreover, BIM and AI integration in construction scheduling enhances the accuracy of cost estimation and budgeting. AI algorithms can analyze historical cost data and project specifications to generate accurate cost estimates for various project components. By considering factors such as material prices, labor costs, and equipment expenses, AI can develop detailed and reliable cost estimates. This accuracy in cost estimation helps project managers develop realistic budgets and allocate resources effectively. Moreover, AI can continuously monitor project expenses in real time, comparing them to the budget and providing alerts if costs exceed the planned expenditures. This proactive cost management ensures that the project stays within budget constraints and helps prevent financial overruns. The integration of Building Information Modelling (BIM) and Artificial Intelligence (AI) in construction scheduling represents a paradigm shift in the construction industry [8,9,12]. It revolutionizes the way construction projects are planned, managed, and executed, leading to increased efficiency, reduced delays, minimized costs, and successful and sustainable outcomes. This integration transforms traditional scheduling practices, paving the way for smarter, more efficient, and more environmentally friendly construction projects.

Table 1 Integrating Building Information modelling (BIM) and Artificial Intelligence (AI) for construction schedule, cost, quality, and safety management

Sl. No.	Aspect	Building Information Modeling (BIM)	Artificial Intelligence (AI)	Integration Methods	Challenges	Future Directions
1	Construction Schedule Management	Utilizes 3D models and clash detection for visualizing construction processes.	Employs AI algorithms for accurate project timeline predictions based on historical data.	Real-time adjustments using predictive analytics.	Data interoperability, varying standards.	Enhanced interoperability, IoT integration, dynamic scheduling via AI.
2	Cost Management	Provides precise quantity takeoffs and cost estimation through 3D models.	Utilizes AI for cost analysis from diverse data sources, ensuring accurate forecasts.	Integrates BIM data into AI algorithms for reliable budgeting.	Cost data accuracy, integration complexity.	Advanced cost data validation algorithms, blockchain for transparent tracking.
3	Quality Management	Facilitates 3D visualization for clash detection and coordination among systems.	Employs AI with image recognition and sensors to monitor construction quality.	BIM 3D models serve as references for AI-driven quality control.	Data accuracy, real-time monitoring complexity.	AI-driven automated quality assurance, advanced real-time sensor technologies.
4	Safety Management	Includes safety-related information such as escape routes in 3D models.	Utilizes AI to predict safety hazards and suggest preventive measures.	BIM data on safety features enhances AI-driven risk assessment and safety protocols.	Data security, privacy, AI reliability.	Enhanced AI algorithms, privacy-preserving techniques, standardized safety data formats.

Virtual Design and Construction (VDC)

Virtual Design and Construction (VDC) extends the capabilities of Building Information Modeling (BIM) by integrating BIM models with project schedules. This integration links 3D models with scheduling data, allowing project managers to visualize the construction process within a simulated environment. Utilizing AI algorithms, VDC enables the analysis of this integrated data to optimize construction sequences. This optimization ensures that tasks are scheduled in the most efficient order. For example, AI can identify opportunities for parallel task execution, thereby reducing the project duration without compromising quality or safety.

Real-time Monitoring and Predictive Analysis

The combination of BIM and AI facilitates real-time monitoring of construction activities. On-site sensors and Internet of Things (IoT) devices collect data, which is fed into the BIM-AI system. AI algorithms analyze this real-time data, predicting potential delays based on historical patterns and current site conditions. Project managers receive timely alerts, allowing them to take corrective actions promptly. This proactive approach helps prevent schedule overruns.

Resource Management and Allocation

AI algorithms optimize resource allocation by considering the skills, availability, and location of workers and equipment. By analyzing BIM data, AI systems assess the requirements of different construction tasks and allocate resources accordingly. For instance, if a specific task requires specialized machinery, AI can identify the nearest available equipment, ensuring timely task completion and adherence to the schedule.

Clash Detection and Conflict Resolution

BIM models play a crucial role in clash detection, identifying conflicts during the design phase. When integrated with AI, these clashes can be automatically analyzed and resolved. AI algorithms assess the impact of clashes on the construction schedule and propose alternative solutions. Swift resolution of design conflicts ensures uninterrupted progress in the construction process, maintaining the project timeline.

Integrating BIM and AI for smart construction cost management

In recent years, the construction industry has witnessed remarkable progress, because of transformative technologies like Building Information Modeling (BIM) and Artificial Intelligence (AI) [5,7,11]. These innovations have reshaped how construction projects are conceived, executed, and overseen. When seamlessly integrated, BIM and AI create a powerful synergy that revolutionizes construction cost management. The synergy of BIM and AI in construction cost management amplifies their respective benefits. BIM lays the groundwork by providing a detailed digital representation of the project, while AI augments this representation with intelligent analysis and predictive capabilities [5,6]. A pivotal integration point involves using AI algorithms to analyze BIM data. The rich information in the BIM model, including geometric data, material specifications, and project schedules, serves as input for AI algorithms [8,11]. These algorithms can scrutinize this data to identify cost-related patterns and correlations [21-26]. For instance, AI can analyze historical BIM data from similar projects to identify common cost drivers and factors influencing project expenses [23,24]. Armed with this knowledge, construction professionals can make more informed decisions when estimating costs for new projects [27-30].

Furthermore, AI enhances BIM-based quantity takeoffs by automatically extracting relevant information from the BIM model [21,22]. AI processes the BIM data, identifying elements, calculating quantities, and cross-referencing this information with historical cost data. This integrated approach ensures that quantity takeoffs are not only accurate but also informed by historical cost trends, leading to more dependable cost estimates. In addition, AI-powered tools enable real-time cost monitoring during construction. By integrating BIM data with AI algorithms, construction companies can monitor project progress and expenses in real-time [5,19,20]. AI analyzes actual construction data against the planned budget and schedule, identifying discrepancies and potential cost overruns. This real-time monitoring facilitates proactive decision-making, allowing project managers to promptly address issues and maintain financial control.

Future Outlook:

The integration of BIM and AI for construction cost management represents a paradigm shift in the industry [5,6]. As AI algorithms become more sophisticated and capable of handling large and complex datasets, their applications in construction cost management will continue to expand. Future developments may include AI-driven predictive analytics that forecast project costs with unprecedented accuracy, enabling construction companies to optimize budgeting and resource allocation. Advancements in machine learning and natural language processing could lead to AI-powered virtual assistants tailored for construction professionals. These virtual assistants could analyze BIM data, answer cost-related queries, and provide real-time insights, enhancing decision-making and collaboration among project teams.

By harnessing the detailed digital representation of BIM and the analytical power of AI algorithms, construction companies can achieve more accurate cost estimates, streamline quantity takeoffs, and monitor project expenses in real-time. While challenges exist, ongoing technological advancements and increased industry awareness are driving the construction sector toward a future where AI-enhanced BIM processes become the norm, enabling more efficient and cost-effective project completion [31-35]. Table 2 shows the integration of BIM and AI for cost management.

Table 2 Integration of BIM and AI for cost management

Sl. No.	**Type of Cost**	**Role of Building Information Modelling (BIM)**	**Role of Artificial Intelligence (AI)**	**Integration of BIM and AI**
1	Material Costs	BIM ensures accurate material estimation through 3D models and databases.	AI analyzes historical material data and market trends for predictive cost analysis.	BIM data is combined with AI algorithms to optimize procurement, predicting material needs and costs efficiently.
2	Labor Costs	BIM aids in efficient labor allocation and project visualization.	AI estimates labor costs based on historical data and project parameters.	BIM schedules are analyzed by AI, predicting labor requirements and optimizing workforce allocation.
3	Equipment Costs	BIM assists in equipment planning with spatial considerations.	AI predicts equipment breakdowns and optimizes usage through performance data.	BIM's equipment data integrates with AI algorithms for optimal usage, reducing downtime and repair expenses.
4	Design and Planning Costs	BIM streamlines collaborative design processes and provides detailed visualizations.	AI automates design evaluations and generates cost-efficient alternatives.	AI analyzes BIM designs, automating evaluations for streamlined and cost-effective planning.
5	Construction Costs	BIM enables 3D modeling and clash detection.	AI monitors real-time project data, identifying issues and deviations.	BIM models integrate with AI for real-time monitoring, reducing rework and construction-related costs.
6	Operation and Maintenance Costs	BIM offers detailed asset data for maintenance planning.	AI predicts maintenance needs and optimizes operational efficiency.	BIM's asset data combines with AI algorithms, ensuring proactive maintenance and cost-effective operations.

Following section delves into the symbiotic relationship between BIM and AI in managing construction costs, focusing on different cost categories and elucidating how this integration can be harnessed for efficient project delivery.

1. Material Cost Management:

Efficient management of construction costs inherently revolves around materials. By combining AI algorithms with BIM, historical material prices, market trends, and supplier data can be analyzed [36-40]. Predictive algorithms forecast material costs, empowering project managers to make informed procurement decisions. AI-driven systems identify optimal suppliers based on cost, quality, and delivery time, ensuring materials are sourced at the best prices. Simultaneously, BIM offers a 3D visualization of the project, facilitating precise quantity take-offs and material estimation. The integration of AI with BIM enables real-time tracking of material usage, preventing over-ordering or shortages and optimizing material costs.

2. Labor Cost Management:

Effectively managing labor costs in construction projects necessitates intricate analysis of factors like skill levels, wages, and project timelines. AI-powered analytics scrutinize labor data from past projects, considering variables such as skillsets, productivity, and overtime patterns. Utilizing machine learning algorithms, AI predicts labor requirements based on project specifications, ensuring the deployment of the right number of skilled workers at the right time. BIM complements this process by creating detailed 3D models, allowing simulations of construction processes. AI algorithms analyze these simulations, optimizing labor workflows, reducing idle time, and enhancing productivity. AI-driven tools monitor worker performance, identifying areas for improvement and training, ultimately augmenting labor efficiency and reducing costs.

3. Overhead Cost Management:

Overhead costs, including administrative expenses, utilities, and equipment maintenance, pose challenges in construction projects. AI-driven predictive analytics analyze historical overhead cost data, identifying patterns and anomalies. Project managers can make data-driven decisions to optimize overhead expenses based on these patterns. AI-powered systems automate administrative tasks, reducing the need for extensive manual labor and minimizing overhead costs associated with paperwork and documentation. Integrated with AI, BIM provides real-time insights into equipment usage and maintenance schedules. Predictive maintenance algorithms anticipate equipment failures, enabling proactive maintenance and minimizing downtime, thereby lowering overhead costs related to equipment repair and replacement.

4. Time Management and Project Delays:

Delays in construction projects can significantly impact costs. AI algorithms analyze historical project data, weather patterns, and other external factors to predict potential delays. By identifying risk factors, project managers can develop contingency plans and allocate resources effectively. BIM, in conjunction with AI, enables 4D simulations, incorporating the element of time into the project's 3D model. This allows project managers to visualize the construction process over time, identifying bottlenecks and potential delays before they occur. By optimizing the construction sequence through AI-driven simulations, project timelines can be streamlined, reducing the risk of delays and associated costs.

5. Quality Control and Rework Costs:

Maintaining high-quality standards is crucial in construction projects to avoid rework costs and ensure customer satisfaction. AI-powered computer vision systems analyze BIM-generated 3D models and on-site images to identify defects and deviations from the original design [41-43]. By detecting issues in real-time, project managers can address them promptly, reducing the likelihood of rework. Moreover, AI algorithms assess the quality of workmanship by analyzing images and sensor data, ensuring that construction standards are met. By minimizing rework and ensuring high-quality construction, costs associated with rectifying defects and customer complaints are significantly reduced.

6. Risk Management and Cost Contingency:

Construction projects inherently involve risks related to weather, regulatory changes, and unforeseen site conditions. AI-driven risk assessment models analyze historical data and external factors to identify potential risks. By quantifying these risks, project managers can allocate appropriate contingencies in the budget, preventing cost overruns. BIM complements this process by providing detailed site information, enabling AI algorithms to assess site-specific risks accurately. Additionally, AI-powered predictive analytics continuously monitor project variables, updating risk assessments in real-time. This dynamic approach to risk management ensures that cost contingencies are adjusted promptly based on changing project conditions, enhancing the overall accuracy of cost estimation and control.

Integrating Building Information Modelling (BIM) and Artificial Intelligence (AI) for construction cost management represents a transformative approach in the construction industry. By leveraging the power of AI algorithms and the detailed insights provided by BIM, construction projects can be optimized across various cost categories. Material costs can be streamlined through predictive analytics, ensuring optimal procurement and usage. Labor costs can be managed efficiently by predicting requirements and optimizing workflows. Overhead costs can be minimized through automation and predictive maintenance. Project delays can be mitigated through predictive analysis and 4D simulations. Quality control can be enhanced, reducing rework costs and ensuring customer satisfaction. Risk management can be proactive, enabling accurate cost contingencies. This integration not only enhances cost management but also improves overall project efficiency, accuracy, and sustainability. As technology continues to advance, the synergy between BIM and AI will play an increasingly vital role in shaping the future of the construction industry, leading to more successful, cost-effective, and timely construction projects. Embracing such integration is not just a choice but a necessity for companies striving to stay competitive in today's rapidly evolving technological landscape [37,43-48].

Integrating BIM and AI for smart construction quality management

In the ever-changing realm of construction, the convergence of Building Information Modeling (BIM) and Artificial Intelligence (AI) emerges as a transformative force, offering unprecedented advancements across various facets of the industry [5,6,18,20]. This integration holds remarkable potential, particularly in the domain of construction quality management [41,42,49-54]. The fusion of BIM and AI introduces an innovative approach that harnesses the power of data, automation, and intelligent algorithms to enhance quality management techniques, ensuring superior results and streamlined processes. When BIM and AI converge, they create a symbiotic alliance that transforms construction quality management [50,52]. This integration facilitates the seamless exchange of data and insights between BIM models and AI algorithms, fostering an ecosystem where information is converted into actionable intelligence.

Predictive Analytics and Risk Management

AI algorithms, when applied to historical project data stored within BIM models, can predict potential risks and challenges. By analyzing past project performance, AI can identify patterns and correlations, enabling the anticipation of issues that might impact quality. Predictive analytics not only forewarns about potential problems but also suggests proactive measures, allowing construction teams to mitigate risks before they escalate. For instance, AI algorithms can analyze past project delays and their causes, identify patterns related to quality issues, and predict potential bottlenecks in the construction process. This predictive capability empowers project managers to allocate resources judiciously, plan schedules effectively, and implement quality assurance measures preemptively.

Automated Quality Inspections

Traditionally, quality inspections are conducted manually, requiring significant time and resources. BIM integrated with AI enables automated quality inspections through techniques like computer vision. High-resolution images and laser scans captured during construction are processed using AI algorithms, which can identify defects, deviations from the design, and structural issues with remarkable accuracy. These algorithms, trained on vast datasets, learn to recognize common quality issues such as cracks, misalignments, or imperfections. By automating the inspection process, construction teams can expedite quality assessments, ensuring that deviations from the design are promptly identified and rectified. This not only enhances the speed of inspections but also significantly improves their accuracy, minimizing the likelihood of overlooking critical issues.

Design Optimization and Simulation

BIM, enriched by AI algorithms, facilitates design optimization and simulation techniques that enhance construction quality. AI-driven generative design algorithms explore countless design permutations based on specified criteria such as material efficiency, cost-effectiveness, and structural integrity. These algorithms iteratively generate and evaluate designs, considering an array of factors that influence quality. By simulating various design scenarios, construction teams can assess the potential impact on quality during the planning phase itself. For instance, simulations can analyze how different structural configurations withstand environmental stresses or how variations in material choices affect durability. This proactive approach ensures that the final design aligns with the highest quality standards, minimizing the need for costly modifications during the construction phase.

Supply Chain Optimization

The integration of BIM and AI extends beyond the construction site to optimize the construction supply chain. AI algorithms can analyze supply chain data within BIM models, predicting demand patterns, optimizing inventory levels, and identifying reliable suppliers. By ensuring the timely availability of high-quality materials, construction projects are less susceptible to delays caused by material shortages or inferior products. Furthermore, AI-driven supply chain optimization enhances quality management by enabling traceability and quality assurance across the entire supply chain. Construction materials can be tracked from their source to their final destination, ensuring that they meet the requisite quality standards at every stage of the process. This end-to-end visibility minimizes the risk of substandard materials entering the construction site, thereby upholding the overall quality of the project.

Real-time Monitoring and Quality Control

BIM and AI synergize to enable real-time monitoring and quality control through IoT (Internet of Things) devices and sensors. These devices, embedded within the construction site and integrated with BIM models, collect a plethora of data, ranging from temperature and humidity to structural stress and equipment performance. AI algorithms process this real-time data, identifying deviations from the expected norms and patterns that could indicate quality issues. For instance, sensors detecting variations in concrete curing temperatures can alert construction teams to potential strength deficiencies. Real-time monitoring of structural components can identify deformations or stress concentrations, highlighting areas that require immediate attention. By enabling proactive quality control measures, real-time monitoring ensures that construction teams can address issues promptly, preventing them from escalating into major quality concerns.

Collaborative Decision-making and Communication

Effective communication and collaborative decision-making are pivotal to construction quality management. BIM serves as the central repository for project data, and AI algorithms enhance this functionality by facilitating intelligent data analysis and decision support. Natural language processing (NLP) algorithms, a subset of AI, enable BIM systems to comprehend and respond to human language. This capability enhances communication between stakeholders, allowing project managers, architects, engineers, and contractors to interact with BIM systems conversationally. Queries related to project specifications, quality standards, or design intent can be answered in real-time, fostering a collaborative environment where decisions are made based on accurate and up-to-date information. This streamlined communication enhances the clarity of instructions and expectations, reducing the likelihood of misunderstandings that could compromise construction quality.

This synergy of BIM and AI not only elevates the quality of construction but also optimizes resource utilization, reduces costs, and accelerates project timelines [25.42]. As the construction industry continues to embrace this innovative amalgamation, it is imperative for stakeholders to address challenges proactively, invest in the requisite training and infrastructure, and uphold ethical principles, ensuring that the potential of BIM and AI is harnessed to its fullest, ushering in a new era of excellence in construction quality management.

Total Quality Management (TQM):

Total Quality Management places a strong emphasis on continuous improvement and customer satisfaction. The integration of Building Information Modelling (BIM) and Artificial Intelligence (AI) enables the construction teams to promptly identify quality issues through real-time data analysis. Through the use of automated defect detection and predictive analytics, TQM principles can be effectively put into practice. Data-driven insights derived from BIM and AI empower informed decision-making, ensuring that construction processes are continuously refined to achieve optimal quality outcomes.

Six Sigma:

The Six Sigma methodology aims to minimize defects and enhance processes. AI algorithms can efficiently process extensive data from BIM models to detect variations and deviations. By employing Six Sigma techniques, construction teams can analyze these variations, pinpoint their root causes, and implement corrective actions. AI-powered analytics facilitate data-driven problem-solving, ensuring that processes are honed to achieve the desired quality levels.

Quality Function Deployment (QFD):

QFD translates customer requirements into specific product features. BIM serves as a rich repository of project data, encompassing design specifications and customer preferences. AI algorithms can analyze this data to effectively align construction processes with customer requirements. By integrating customer preferences from BIM into AI algorithms, construction teams can prioritize features and elements that align with customer expectations, thereby enhancing overall project quality.

Statistical Process Control (SPC):

SPC involves monitoring and controlling processes using statistical methods. IoT sensors, in conjunction with BIM and AI, provide real-time data for SPC analysis. Sensors collect data on various parameters, which AI algorithms analyze. Statistical techniques are then applied to assess process stability and identify potential deviations. Construction teams can take immediate corrective actions when deviations are detected, ensuring that processes remain within statistical control and, consequently, maintain high-quality standards.

Plan-Do-Check-Act (PDCA) Cycle:

The PDCA cycle is a systematic approach to quality improvement. BIM and AI support each stage of the PDCA cycle. During the Planning phase, AI-driven predictive analytics assist in setting realistic quality goals based on historical data and project requirements. In the Doing phase, BIM provides a comprehensive view of the construction process, while AI algorithms monitor real-time progress. The Check phase involves analyzing data from BIM and AI to assess whether the project meets quality standards. Finally, in the Act phase, insights gained from BIM and AI are utilized to implement necessary changes, ensuring continuous quality improvement throughout the construction process.

ISO 9000 Standards:

ISO 9000 standards center around quality management systems. BIM acts as a central repository for project documentation, ensuring that all stakeholders have access to standardized and updated information. AI algorithms can conduct audits by comparing project activities and outcomes against ISO standards. Any deviations can be flagged, and corrective actions can be taken promptly. This integration guarantees that construction projects adhere to international quality management standards, promoting consistency and reliability in project delivery.

Integrating BIM and AI for smart construction safety management

The construction industry plays a vital role in global economic development but is also inherently risky, posing numerous safety threats to both workers and the public. Ensuring safety on construction sites is of utmost importance, and emerging technologies offer groundbreaking solutions. Building Information Modelling (BIM) and Artificial Intelligence (AI) stand out as powerful technologies that, when combined, can transform construction safety management [55-59]. BIM serves as a digital representation of a building's physical and functional aspects, fostering collaboration and efficiency in construction [8,9]. AI, on the other hand, empowers machines to perform tasks requiring human-like intelligence, such as learning and problem-solving [11,12]. Integrating such technologies holds the potential to enhance safety protocols, mitigate risks, and ultimately save lives in various sectors [4,19,59-65].

1. Risk Assessment and Predictive Analytics:

By integrating BIM with AI algorithms, vast amounts of data from construction projects can be analyzed to identify potential safety hazards. AI algorithms, utilizing historical and real-time data from BIM models, predict risks and accidents. For example, AI algorithms can analyze clash points and hazardous areas in the BIM model, enabling construction teams to proactively implement safety measures, thereby preventing accidents before they occur. Predictive analytics powered by AI significantly reduce workplace accidents, making construction sites safer for workers.

2. Real-time Monitoring and Safety Alerts:

The integration of BIM and AI allows for real-time monitoring of construction sites using IoT devices. Sensors and cameras, incorporated into the BIM model, collect real-time data about site conditions and worker movements. AI processes this data instantly, detecting unsafe behaviors or conditions. AI algorithms trigger safety alerts in real-time, notifying site supervisors or workers for immediate intervention, thus preventing accidents. Real-time monitoring and AI-driven safety alerts enhance situational awareness and enable rapid response, improving overall safety on construction sites.

3. Virtual Reality (VR) and Simulation for Safety Training:

VR simulations, integrated with BIM and AI, offer immersive safety training experiences for construction workers. AI analyzes the BIM model and creates realistic VR simulations mirroring the construction site environment. Workers undergo safety training in a virtual environment, encountering simulated hazardous situations and learning effective responses. AI customizes these simulations based on project-specific BIM data, ensuring relevant and tailored training scenarios. VR simulations, enhanced by AI-driven interactions, provide a controlled environment for workers to practice safety protocols, improving their preparedness for real-life situations.

4. Automated Safety Inspections and Compliance Monitoring:

Integrating BIM with AI enables automated safety inspections. AI algorithms analyze the BIM model, comparing it against safety regulations and standards. Computer vision technology allows AI to identify safety violations, such as improper scaffolding, in images and videos from construction sites. Automated drones equipped with cameras capture site images, processed by AI algorithms to assess safety compliance. Automation ensures prompt identification and rectification of safety violations, reducing the risk of accidents.

5. Predictive Maintenance for Equipment and Infrastructure:

AI integrated with BIM enables predictive maintenance for construction equipment. Sensors on machinery collect data on performance metrics. AI algorithms analyze this data to predict equipment failures, allowing proactive maintenance and minimizing accidents caused by equipment failures. Predictive maintenance extends equipment lifespan, ensuring safe and efficient operation throughout construction projects.

6. Enhanced Collaboration and Communication:

BIM serves as a centralized platform for sharing project information. AI enhances collaboration by processing and organizing this data. Natural Language Processing (NLP) algorithms interpret and generate human-like text, facilitating seamless communication among team members. AI-powered chatbots provide instant responses to safety-related queries, ensuring access to relevant safety information. AI analyzes communication patterns, addressing misunderstandings or conflicts promptly, fostering a collaborative environment for effective safety management.

7. Data-driven Decision Making:

Integrating BIM and AI generates valuable data related to construction processes and safety incidents. AI processes large datasets, identifying patterns and trends related to safety incidents. Construction companies use these insights for data-driven decision-making, prioritizing safety measures based on significant risks identified through AI analysis. Data-driven decision-making ensures efficient resource allocation, enhancing safety on construction sites. Table 3 shows the Integration of Building Information Modelling (BIM) and Artificial Intelligence (AI) for construction safety management.

Table 3 Integrating Building Information Modelling (BIM) and Artificial Intelligence (AI) for construction safety management

Sl. No.	Safety Management Aspect	Description	Key Technologies	Implementation
1	Risk Prediction	AI algorithms analyze historical data to predict safety risks, enhancing proactive risk management.	Machine Learning, Predictive Analytics	Analyze past incidents to forecast potential risks; proactive risk mitigation.
2	Site Monitoring	Real-time monitoring using IoT devices and AI-powered cameras identifies hazards promptly for immediate action.	IoT, Computer Vision	Instant alerts for hazardous conditions; real-time monitoring of safety zones.
3	Safety Training	VR-based training modules utilize BIM data, enabling workers to simulate real construction scenarios for safety training.	Virtual Reality (VR), BIM	Immersive training based on authentic site data; enhance safety awareness.
4	Clash Detection	BIM identifies clashes in building elements, preventing accidents arising from design conflicts.	BIM, Clash Detection Tools	Early detection and resolution of design conflicts; seamless construction.
5	Resource Allocation	AI optimizes resource allocation ensuring efficient deployment of safety equipment and personnel.	AI, Optimization Algorithms	Resource allocation based on real-time needs; maximize safety measures.
6	Emergency Response	Intelligent systems integrate BIM and AI to guide workers to safety during emergencies using geolocation technology.	BIM, AI, Geolocation Technology	Real-time guidance to safe zones; enhance emergency response efficiency.
7	Predictive Maintenance	AI-driven predictive maintenance minimizes accidents by identifying equipment issues before failures occur.	AI, Predictive Maintenance Tools	Preventive maintenance based on equipment health analysis; reduce equipment risks.
8	Automated Documentation	AI automates documentation, ensuring accurate recording of safety protocols, incident reports, and compliance data.	AI, Natural Language Processing (NLP)	Automatic generation of reports; streamline documentation processes.
9	Behavior Analysis	AI-powered cameras analyze worker behavior, providing real-time feedback on unsafe actions to enhance safety awareness.	AI, Computer Vision	Recognize and address unsafe actions; immediate feedback for behavioral improvements.
10	Safety Compliance Monitoring	AI monitors construction activities for adherence to safety regulations, ensuring real-time compliance and reducing risks.	AI, Compliance Monitoring Tools	Continuous monitoring for real-time adherence; minimize compliance risks.
11	Data Analytics for Safety	Analyze BIM and AI-generated data to identify patterns,	Data Analytics, BIM, AI	Identify safety trends for proactive safety

		enabling proactive safety measures based on trends and insights.		management; data-driven decision-making.
12	Remote Inspections	AI-assisted drones with cameras conduct remote inspections, reducing the need for physical presence in hazardous areas.	AI, Drones, Remote Sensing Technologies	Remote inspections for hazardous sites; enhance safety for on-site personnel.

Challenges of integrating BIM and AI for smart construction schedule, cost, quality, and safety management

BIM serves as a digital representation of a building's or infrastructure's physical and functional characteristics, providing a shared knowledge resource. AI, on the other hand, involves the simulation of human intelligence in machines programmed to think and learn like humans. The integration of BIM and AI in construction projects offers the promise of improved efficiency and effectiveness in managing schedules, costs, quality, and safety [19,20,25,66-70]. However, such integration is not without its challenges [5,70-75].

Data Integration and Interoperability:

A primary challenge in combining BIM and AI for smart construction management is seamlessly integrating and ensuring data interoperability. BIM generates extensive data related to building attributes, spatial relationships, geographic information, and quantities. AI systems rely on diverse and well-structured data to make intelligent decisions. Bridging the gap between BIM and AI often encounters difficulties due to differences in data formats, standards, and semantics. The development of standardized approaches for data representation and sharing is crucial to address such challenge.

Complexity and Scalability:

Construction projects inherently involve complexity with multiple stakeholders, diverse materials, intricate processes, and dynamic environments. Integrating AI with BIM to manage this complexity and ensure scalability is a significant challenge. AI algorithms must handle large and diverse datasets, making real-time decisions to optimize various aspects of the project. The dynamic nature of construction projects adds complexity to AI models, requiring the development of algorithms that can adapt to different project scales while maintaining accuracy and efficiency.

Algorithm Development and Training:

Creating AI algorithms tailored to the construction industry demands a deep understanding of construction processes and domain-specific knowledge. Training AI models to recognize patterns and make predictions related to construction schedules, costs, quality, and safety requires extensive datasets, which are often limited in the construction domain. The dynamic nature of construction projects further complicates matters, as AI models need continuous training and refinement to improve their accuracy and reliability.

Ethical and Legal Concerns:

Integrating AI into construction raises ethical and legal concerns surrounding data privacy, security, and liability. Construction projects involve sensitive information related to designs, budgets, schedules, and safety protocols. AI algorithms processing this data must adhere to strict ethical guidelines to ensure data confidentiality and privacy. Determining liability in cases of AI-generated errors or decisions presents a legal challenge that needs to be addressed. Clear regulations and standards governing the ethical and legal aspects of AI integration in construction are necessary to mitigate these concerns.

User Acceptance and Training:

The successful integration of BIM and AI in construction projects relies on the acceptance and understanding of these technologies by industry professionals. Key stakeholders, including architects, engineers, contractors, and project managers, need effective training to use BIM and AI tools efficiently. Resistance to change, lack of awareness, and insufficient training programs are common challenges during the implementation of new technologies. Building a proficient workforce in both BIM and AI applications is essential for realizing the full potential of these technologies in construction management.

Real-time Data Processing and Decision-making:

Construction projects operate in real-time, with decisions made on the spot to address emerging challenges and ensure project progression. Integrating AI with BIM for real-time data processing and decision-making requires high-speed data processing capabilities. Delays in data processing can lead to misinformed decisions, impacting project schedules, costs, quality, and safety. Implementing AI algorithms capable of processing large volumes of data in real-time and providing immediate actionable insights is a technical challenge faced by the construction industry in the integration process.

Cost Implications and Return on Investment:

Integrating BIM and AI technologies in construction projects involves significant costs related to software licenses, hardware infrastructure, training, and implementation. Construction companies must carefully weigh these costs against the potential benefits and return on investment (ROI). Demonstrating the tangible benefits of AI integration, such as reduced construction time, cost savings, improved quality, and enhanced safety, is essential to justify the initial investment. Calculating and ensuring a positive ROI within a reasonable timeframe is a challenge faced by construction firms considering BIM and AI integration.

Cybersecurity Risks:

The integration of BIM and AI introduces cybersecurity risks, especially when sensitive construction data is stored and processed digitally. Construction projects are attractive targets for cyberattacks due to the valuable information they contain. AI algorithms analyzing BIM data are vulnerable to hacking, data breaches, and unauthorized access. Ensuring robust cybersecurity measures to protect BIM and AI systems from malicious activities is a critical challenge. Construction companies must invest in solutions to safeguard their digital assets and maintain the integrity of project data [76-81].

Conclusions

The amalgamation of Building Information Modelling (BIM) and Artificial Intelligence (AI) within smart construction practices signifies a crucial advancement in the construction field. Our study delves into the challenges and potential directions of this integration, underscoring its accomplishments and the challenges that demand attention for successful implementation. Undeniably, the integration of BIM and AI has brought about a transformative shift in the planning, execution, and supervision of construction projects. Leveraging AI algorithms and machine learning, stakeholders now benefit from improved predictive analytics, intelligent decision-making, and streamlined workflows. By combining BIM's digital representations with AI's data processing abilities, project managers can foresee scheduling conflicts, optimize resource allocation, and enhance overall project efficiency. Our research identified data integration complexity and interoperability as a key challenge. The diverse software and data formats used by different stakeholders necessitate seamless data exchange.

Standardization efforts are vital to enable effective communication between BIM and AI technologies, fostering a cohesive and collaborative construction environment. Moreover, adopting BIM and AI necessitates a shift in the industry's workforce, demanding proficiency in both construction and digital technologies. Comprehensive training programs are crucial to equip professionals with the required expertise. Encouraging interdisciplinary collaboration is vital to fostering a holistic understanding of these integrated technologies. Additionally, robust data governance and security measures are essential to protect sensitive project information. Emphasizing the development of data governance frameworks and cybersecurity protocols is crucial to mitigate potential risks associated with data breaches and unauthorized access. Looking ahead, collaborative research and development efforts are imperative. Advancements in AI algorithms and BIM software will lead to sophisticated predictive models and decision support systems tailored to construction needs. Embracing emerging technologies such as the Internet of Things (IoT) and augmented reality (AR) will amplify integrated BIM and AI systems' capabilities.

Furthermore, fostering an ecosystem of open data exchange and interoperability standards will enhance the scalability and adaptability of these solutions. Encouraging industry-wide collaboration and partnerships will accelerate their widespread adoption, propelling the construction industry towards efficiency, sustainability, and innovation. The integration of BIM and AI promises to revolutionize the construction industry, offering opportunities for improved project management, cost optimization, quality assurance, and safety enhancement. Addressing challenges through collaborative efforts can pave the way for the seamless integration and widespread adoption of these transformative technologies. A forward-thinking approach can harness the full potential of integrated BIM and AI systems, shaping a smarter, more efficient, and sustainable future for construction project delivery and management.

References

[1] Sacks, R., Girolami, M., & Brilakis, I. (2020). Building information modelling, artificial intelligence and construction tech. Developments in the Built Environment, 4, 100011.

[2] Zhang, F., Chan, A. P., Darko, A., Chen, Z., & Li, D. (2022). Integrated applications of building information modeling and artificial intelligence techniques in the AEC/FM industry. Automation in Construction, 139, 104289.

[3] Locatelli, M., Seghezzi, E., Pellegrini, L., Tagliabue, L. C., & Di Giuda, G. M. (2021). Exploring natural language processing in construction and integration with building information modeling: A scientometric analysis. Buildings, 11(12), 583.

[4] Yin, X., Liu, H., Chen, Y., & Al-Hussein, M. (2019). Building information modelling for off-site construction: Review and future directions. Automation in construction, 101, 72-91.

[5] Darko, A., Chan, A. P., Yang, Y., & Tetteh, M. O. (2020). Building information modeling (BIM)-based modular integrated construction risk management–Critical survey and future needs. Computers in Industry, 123, 103327.

[6] Ozturk, G. B., & Yitmen, I. (2019, February). Conceptual model of building information modelling usage for knowledge management in construction projects. In IOP Conference Series: Materials Science and Engineering (Vol. 471, No. 2, p. 022043). IOP Publishing.

[7] Bui, N., Merschbrock, C., & Munkvold, B. E. (2016). A review of Building Information Modelling for construction in developing countries. Procedia Engineering, 164, 487-494.

[8] Crotty, R. (2013). The impact of building information modelling: transforming construction. Routledge.

[9] Cao, D., Wang, G., Li, H., Skitmore, M., Huang, T., & Zhang, W. (2015). Practices and effectiveness of building information modelling in construction projects in China. Automation in construction, 49, 113-122.

[10] Pan, Y., & Zhang, L. (2021). Roles of artificial intelligence in construction engineering and management: A critical review and future trends. Automation in Construction, 122, 103517.

[11] Zhang, L., Pan, Y., Wu, X., & Skibniewski, M. J. (2021). Artificial intelligence in construction engineering and management (pp. 95-124). Singapore:: Springer.

[12] Eber, W. (2020). Potentials of artificial intelligence in construction management. Organization, technology & management in construction: an international journal, 12(1), 2053-2063.

[13] Abioye, S. O., Oyedele, L. O., Akanbi, L., Ajayi, A., Delgado, J. M. D., Bilal, M., ... & Ahmed, A. (2021). Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges. Journal of Building Engineering, 44, 103299.

[14] Liu, N., Kang, B. G., & Zheng, Y. (2018, November). Current trend in planning and scheduling of construction project using artificial in telligence. In IET Doctoral Forum on Biomedical Engineering, Healthcare, Robotics and Artificial Intelligence 2018 (BRAIN 2018) (pp. 1-6). IET.

[15] Nusen, P., Boonyung, W., Nusen, S., Panuwatwanich, K., Champrasert, P., & Kaewmoracharoen, M. (2021). Construction planning and scheduling of a renovation project using BIM-based multi-objective genetic algorithm. Applied Sciences, 11(11), 4716.

[16] Wang, H., & Hu, Y. (2022). Artificial Intelligence Technology Based on Deep Learning in Building Construction Management System Modeling. Advances in Multimedia, 2022.
[17] Abioye, S. O., Oyedele, L. O., Akanbi, L., Ajayi, A., Delgado, J. M. D., Bilal, M., ... & Ahmed, A. (2021). Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges. Journal of Building Engineering, 44, 103299.
[18] Doukari, O., Seck, B., & Greenwood, D. (2022). The creation of construction schedules in 4D BIM: a comparison of conventional and automated approaches. Buildings, 12(8), 1145.
[19] Singh, J., & Anumba, C. J. (2022). Real-time pipe system installation schedule generation and optimization using artificial intelligence and heuristic techniques. Journal of Information Technology in Construction, 27.
[20] Abanda, F. H., Musa, A. M., Clermont, P., Tah, J. H., & Oti, A. H. (2020). A BIM-based framework for construction project scheduling risk management. International Journal of Computer Aided Engineering and Technology, 12(2), 182-218.
[21] Igwe, U. S., Mohamed, S. F., & Azwarie, M. B. M. D. (2020, July). Recent Technologies in Construction; A Novel Search for Total Cost Management of Construction Projects. In IOP Conference Series: Materials Science and Engineering (Vol. 884, No. 1, p. 012041). IOP Publishing.
[22] Robson, A., Boyd, D., & Thurairajah, N. (2016). Studying 'cost as information'to account for construction improvements. Construction Management and Economics, 34(6), 418-431.
[23] Xu, S., Liu, K., & Tang, L. C. (2013). Cost estimation in building information models. In ICCREM 2013: Construction and Operation in the Context of Sustainability (pp. 555-566).
[24] Seidu, R. D., Young, B. E., Clack, J., Adamu, Z., & Robinson, H. (2020). Innovative changes in quantity surveying practice through BIM, big data, artificial intelligence and machine learning. Applied Science University Journal of Natural Science., 4(1), 37-47.
[25] Doukari, O., Kassem, M., & Greenwood, D. (2023). Building Information Modelling. In Disrupting Buildings: Digitalisation and the Transformation of Deep Renovation (pp. 39-51). Cham: Springer International Publishing.
[26] Khan, A., Ghadg, A. N., & Nagrale, D. P. P. (2018). Evaluating Benefits of Building Information Modelling (BIM) Using A 5D Model for Construction Project. Available at SSRN 3371434.
[27] Rane, N. L., & Attarde, P. M. (2016). Application of value engineering in commercial building projects. International Journal of Latest Trends in Engineering and Technology, 6(3), 286-291.
[28] Rane, N. L., (2016). Application of value engineering techniques in building construction projects. International Journal of Engineering Sciences & Technology, 5(7).
[29] Patil, D. R., Rane, N. L., (2023) Customer experience and satisfaction: importance of customer reviews and customer value on buying preference, International Research Journal of Modernization in Engineering Technology and Science, 5(3), 3437- 3447. https://www.doi.org/10.56726/IRJMETS36460
[30] Rane, N. L., (2016) Application of value engineering techniques in construction projects, international journal of engineering sciences & research technology, 5(7), 1409-1415. https://doi.org/10.5281/zenodo.58597
[31] Ghaffarianhoseini, A., Tookey, J., Ghaffarianhoseini, A., Naismith, N., Azhar, S., Efimova, O., & Raahemifar, K. (2017). Building Information Modelling (BIM) uptake: Clear benefits, understanding its implementation, risks and challenges. Renewable and sustainable energy reviews, 75, 1046-1053.
[32] Kelly, C. J., Karthikesalingam, A., Suleyman, M., Corrado, G., & King, D. (2019). Key challenges for delivering clinical impact with artificial intelligence. BMC medicine, 17, 1-9.
[33] Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2019). Artificial intelligence and the public sector—applications and challenges. International Journal of Public Administration, 42(7), 596-615.
[34] Dwivedi, Y. K., Hughes, L., Ismagilova, E., Aarts, G., Coombs, C., Crick, T., ... & Williams, M. D. (2021). Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. International Journal of Information Management, 57, 101994.
[35] Duan, Y., Edwards, J. S., & Dwivedi, Y. K. (2019). Artificial intelligence for decision making in the era of Big Data–evolution, challenges and research agenda. International journal of information management, 48, 63-71.
[36] Yin, X., Liu, H., Chen, Y., & Al-Hussein, M. (2019). Building information modelling for off-site construction: Review and future directions. Automation in construction, 101, 72-91.

[37] Alwan, Z., Jones, P., & Holgate, P. (2017). Strategic sustainable development in the UK construction industry, through the framework for strategic sustainable development, using Building Information Modelling. Journal of cleaner production, 140, 349-358.
[38] Zang, Y., Zhang, F., Di, C. A., & Zhu, D. (2015). Advances of flexible pressure sensors toward artificial intelligence and health care applications. Materials Horizons, 2(2), 140-156.
[39] Wong, K. D., & Fan, Q. (2013). Building information modelling (BIM) for sustainable building design. Facilities, 31(3/4), 138-157.
[40] Crotty, R. (2013). The impact of building information modelling: transforming construction. Routledge.
[41] Ma, G., Wu, M., Wu, Z., & Yang, W. (2021). Single-shot multibox detector-and building information modeling-based quality inspection model for construction projects. Journal of Building Engineering, 38, 102216.
[42] Bruno, S., De Fino, M., & Fatiguso, F. (2018). Historic Building Information Modelling: performance assessment for diagnosis-aided information modelling and management. Automation in Construction, 86, 256-276.
[43] Bassir, D., Lodge, H., Chang, H., Majak, J., & Chen, G. (2023). Application of artificial intelligence and machine learning for BIM. International Journal for Simulation and Multidisciplinary Design Optimization, 14, 5.
[44] Rane, N. L., Achari, A., Choudhary, S. P., Mallick, S. K., Pande, C. B., Srivastava, A., & Moharir, K. (2023). A decision framework for potential dam site selection using GIS, MIF and TOPSIS in Ulhas river basin, India. Journal of Cleaner Production, 138890. https://doi.org/10.1016/j.jclepro.2023.138890
[45] Rane, N. L., Achari, A., Saha, A., Poddar, I., Rane, J., Pande, C. B., & Roy, R. (2023). An integrated GIS, MIF, and TOPSIS approach for appraising electric vehicle charging station suitability zones in Mumbai, India. Sustainable Cities and Society, 104717. https://doi.org/10.1016/j.scs.2023.104717
[46] Gautam, V. K., Pande, C. B., Moharir, K. N., Varade, A. M., Rane, N. L., Egbueri, J. C., & Alshehri, F. (2023). Prediction of Sodium Hazard of Irrigation Purpose using Artificial Neural Network Modelling. Sustainability, 15(9), 7593. https://doi.org/10.3390/su15097593
[47] Moharir, K. N., Pande, C. B., Gautam, V. K., Singh, S. K., & Rane, N. L. (2023). Integration of hydrogeological data, GIS and AHP techniques applied to delineate groundwater potential zones in sandstone, limestone and shales rocks of the Damoh district, (MP) central India. Environmental Research, 115832. https://doi.org/10.1016/j.envres.2023.115832
[48] Rane, N. L., Anand, A., Deepak K., (2023). Evaluating the Selection Criteria of Formwork System (FS) for RCC Building Construction. International Journal of Engineering Trends and Technology, vol. 71, no. 3, pp. 197-205. Crossref, https://doi.org/10.14445/22315381/IJETT-V71I3P220
[49] Lekan, A., Clinton, A., Stella, E., Moses, E., & Biodun, O. (2022). Construction 4.0 Application: Industry 4.0, Internet of Things and Lean Construction Tools' Application in Quality Management System of Residential Building Projects. Buildings, 12(10), 1557.
[50] To, T. H. Q., & Phuong, L. Q. (2021). Applying bim and related technologies for maintenance and quality management of construction assets in Vietnam. International Journal of Sustainable Construction Engineering and Technology, 12(5), 125-135.
[51] Abazid, M., Gökçekuş, H., & Celik, T. (2021). Implementation of TQM and the integration of BIM in the construction management sector in Saudi Arabia validated with hybridized emerging harris hawks optimization (HHO).
[52] Bolpagni, M., Gavina, R., & Ribeiro, D. (Eds.). (2021). Industry 4.0 for the Built Environment: Methodologies, Technologies and Skills (Vol. 20). Springer Nature.
[53] Faraji, A., Rashidi, M., Meydani Haji Agha, T., Rahnamayiezekavat, P., & Samali, B. (2022). Quality Management Framework for Housing Construction in a Design-Build Project Delivery System: A BIM-UAV Approach. Buildings, 12(5), 554.
[54] Chen, S., Zeng, Y., Majdi, A., Salameh, A. A., Alkhalifah, T., Alturise, F., & Ali, H. E. (2023). Potential features of building information modelling for application of project management knowledge areas as advances modeling tools. Advances in Engineering Software, 176, 103372.

[55] Bigham, G. F., Adamtey, S., Onsarigo, L., & Jha, N. (2019). Artificial intelligence for construction safety: Mitigation of the risk of fall. In Intelligent Systems and Applications: Proceedings of the 2018 Intelligent Systems Conference (IntelliSys) Volume 2 (pp. 1024-1037). Springer International Publishing.

[56] Li, R. Y. M., & Li, R. Y. M. (2018). Building Information Modelling and Construction Safety. An Economic Analysis on Automated Construction Safety: Internet of Things, Artificial Intelligence and 3D Printing, 47-72.

[57] Fargnoli, M., & Lombardi, M. (2020). Building information modelling (BIM) to enhance occupational safety in construction activities: Research trends emerging from one decade of studies. Buildings, 10(6), 98.

[58] Collinge, W. H., Farghaly, K., Mosleh, M. H., Manu, P., Cheung, C. M., & Osorio-Sandoval, C. A. (2022). BIM-based construction safety risk library. Automation in Construction, 141, 104391.

[59] Tender, M., Couto, J. P., & Fuller, P. (2022). Improving Occupational Health and Safety Data Integration Using Building Information Modelling: An Initial Literature Review. Occupational and Environmental Safety and Health III, 75-84.

[60] Rane, N. L., Achari, A., Hashemizadeh, A., Phalak, S., Pande, C. B., Giduturi, M., Khan M. Y., Tolche A, D., Tamam, N., Abbas, M,, & Yadav, K. K. (2023). Identification of sustainable urban settlement sites using interrelationship based multi-influencing factor technique and GIS. Geocarto International, 1-27. https://doi.org/10.1080/10106049.2023.2272670

[61] Rane, N., & Jayaraj, G. K. (2021). Stratigraphic modeling and hydraulic characterization of a typical basaltic aquifer system in the Kadva river basin, Nashik, India. Modeling Earth Systems and Environment, 7, 293-306. https://doi.org/10.1007/s40808-020-01008-0

[62] Achari, A., Rane, N. L., Gangar B., (2023). Framework Towards Achieving Sustainable Strategies for Water Usage and Wastage in Building Construction. International Journal of Engineering Trends and Technology, vol. 71, no. 3, pp. 385-394. Crossref, https://doi.org/10.14445/22315381/IJETT-V71I3P241

[63] Rane, N. L., & Jayaraj, G. K. (2022). Comparison of multi-influence factor, weight of evidence and frequency ratio techniques to evaluate groundwater potential zones of basaltic aquifer systems. Environment, Development and Sustainability, 24(2), 2315-2344. https://doi.org/10.1007/s10668-021-01535-5

[64] Rane, N., & Jayaraj, G. K. (2021). Evaluation of multiwell pumping aquifer tests in unconfined aquifer system by Neuman (1975) method with numerical modeling. In Groundwater resources development and planning in the semi-arid region (pp. 93-106). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-68124-1_5

[65] Rane, Nitin (2023) ChatGPT and Similar Generative Artificial Intelligence (AI) for Smart Industry: Role, Challenges and Opportunities for Industry 4.0, Industry 5.0 and Society 5.0. Available at SSRN: https://ssrn.com/abstract=4603234 or http://dx.doi.org/10.2139/ssrn.4603234

[66] Ozturk, G. B., & Yitmen, I. (2019, February). Conceptual model of building information modelling usage for knowledge management in construction projects. In IOP Conference Series: Materials Science and Engineering (Vol. 471, No. 2, p. 022043). IOP Publishing.

[67] Micolier, A., Taillandier, F., Taillandier, P., & Bos, F. (2019). Li-BIM, an agent-based approach to simulate occupant-building interaction from the Building-Information Modelling. Engineering Applications of Artificial Intelligence, 82, 44-59.

[68] Dastbaz, M., Gorse, C., & Moncaster, A. (Eds.). (2017). Building information modelling, building performance, design and smart construction. Springer.

[69] Orooje, M. S., & Latifi, M. M. (2022). A Review of Embedding Artificial Intelligence in Internet of Things and Building Information Modelling for Healthcare Facility Maintenance Management. Energy and Environment Research, 11(2), 1-31.

[70] Lu, Q., Xie, X., Parlikad, A. K., Schooling, J. M., & Konstantinou, E. (2020). Moving from building information models to digital twins for operation and maintenance. Proceedings of the Institution of Civil Engineers-Smart Infrastructure and Construction, 174(2), 46-56.

[71] Rane, Nitin (2023) Contribution and Challenges of ChatGPT and Similar Generative Artificial Intelligence in Biochemistry, Genetics and Molecular Biology. Available at SSRN: https://ssrn.com/abstract=4603219 or http://dx.doi.org/10.2139/ssrn.4603219

[72] Rane, Nitin (2023) Transformers in Material Science: Roles, Challenges, and Future Scope. Available at SSRN: https://ssrn.com/abstract=4609920 or http://dx.doi.org/10.2139/ssrn.4609920

[73] Rane, N. L., Achari, A., & Choudhary, S. P. (2023) enhancing customer loyalty through quality of service: effective strategies to improve customer satisfaction, experience, relationship, and engagement. International Research Journal of Modernization in Engineering Technology and Science, 5(5), 427-452. https://www.doi.org/10.56726/IRJMETS38104
[74] Rane, N. L. (2023). Multidisciplinary collaboration: key players in successful implementation of ChatGPT and similar generative artificial intelligence in manufacturing, finance, retail, transportation, and construction industry. https://doi.org/10.31219/osf.io/npm3d
[75] Rane, N. L., Choudhary, S. P., Tawde, A., & Rane, J. (2023) ChatGPT is not capable of serving as an author: ethical concerns and challenges of large language models in education. International Research Journal of Modernization in Engineering Technology and Science, 5(10), 851-874. https://www.doi.org/10.56726/IRJMETS45212
[76] Raouf, A. M., & Al-Ghamdi, S. G. (2019). Building information modelling and green buildings: Challenges and opportunities. Architectural Engineering and Design Management, 15(1), 1-28.
[77] Yin, X., Liu, H., Chen, Y., & Al-Hussein, M. (2019). Building information modelling for off-site construction: Review and future directions. Automation in construction, 101, 72-91.
[78] Saka, A. B., Chan, D. W. M., & Olawumi, T. O. (2019). A systematic literature review of building information modelling in the architecture, engineering and construction industry-the case of Nigeria.
[79] Dastbaz, M., Gorse, C., & Moncaster, A. (Eds.). (2017). Building information modelling, building performance, design and smart construction. Springer.
[80] Rampini, L., & Cecconi, F. R. (2022). Artificial intelligence in construction asset management: A review of present status, challenges and future opportunities. Journal of Information Technology in Construction, 27, 884-913.
[81] Wang, Z., He, B., Yang, Y., Shen, C., & Peña-Mora, F. (2020, June). Building a next generation AI platform for AEC: A review and research challenges. In Proc. 37th Int. Conf.(CIB) (pp. 27-45).

Declarations

Funding: No funding was received.

Conflicts of interest/Competing interests: No conflict of interest.

Availability of data and material: Not applicable.

Code availability: Not applicable.

Acknowledgements: Not Applicable.